Online Appendix to "The Exchange Rate and the Labor Share: A Material Cost Channel"

Equations, tables and figures without "OA" refer to the paper.

OA.1 Data

A confidential cell is recovered from its section total when it is the only one in its section and year; ten of the 63 sectors still miss some years, which leaves 719 observations. Value added adds the change in stocks of purchased goods to the published figure, $VA = VA^{pub} + \Delta S$. When inflation is high, part of ΔS is a holding gain, so our value added is an upper bound and the published one a lower bound. The services PPI starts in 2017 and is chained back with the section's value added deflator. Only employment activities (N78) have a markup below one (0.992). The markup leaves out the cost of capital, so it is an upper bound on the margin over full cost.

OA.2 The world price instrument

The instrument of eq. (23) prices the 27 goods of Table OA1. The mining product of the 2012 input–output table is split into coal, crude petroleum, natural gas, metal ores and other mining, with a RAS procedure that keeps structural zeros. Six-digit HS codes are assigned to products through the official correspondences from HS to CPC 2.1 and from CPC 2.1 to CPA 2.1. The world price of a good chains the annual log change of its unit values, the mean over its codes weighted by 2013 imports, after changes more than $\ln 2$ from the weighted median are dropped; gold is left out of basic metals. Goods without a world price, such as services, get no weight, so the weights of a sector sum to 0.389 on average.

Two groups of goods carry most of the identification. The Rotemberg weights (Goldsmith-Pinkham et al., 2020) put 43% on waste and scrap and basic metals and 37% on energy products, and almost all of the weight falls on products whose own estimate is negative. Neither group drives the estimate alone: dropping any one product, all energy products or the two largest leaves it between -0.171 and -0.151 (Table OA1). The effective number of products is 15 (Borusyak et al., 2022), and the years 2021 and 2022 carry about 70% of the identifying weight (their share in the covariance between the instrument and the regressor, net of sector and year effects).

OA.3 Further robustness checks

Table OA2 adds two checks to Table B1: a control for material intensity times the real interest rate, and the long difference over 2017–2022. Both leave the estimate of eq. (19) negative and significant; in the long difference the coefficients of the two channels keep their sign but lose precision. Figure OA1 drops one sector at a time.

OA.4 Supplier prices

Table OA3 splits the cumulative responses of eq. (25) by route. Without electricity and gas (Panel B), the network response stops building up after the first month, so the build-up in Table 4 comes mostly from this sector.

Table OA1. Products, world prices and Rotemberg weights of the instrument

Product	Price	Weight, %	Log change, 2020–2022	Rotemberg weight	$\hat{\beta}$ without the product	F
A01 Crops and livestock	UV	1.88	0.41	0.017	−0.157	351
A02 Forestry	UV	0.35	0.53	0.006	−0.159	345
A03 Fish	UV	0.04	0.31	0.000	−0.159	352
B05 Coal	UV	0.27	1.16	0.012	−0.158	345
B06 Crude petroleum	WB	0.00	0.86	0.000	−0.159	352
B06 Natural gas	WB	0.59	2.52	0.126	−0.158	226
B07 Metal ores	UV	0.72	0.40	0.063	−0.159	240
B08 Other mining	UV	0.44	0.44	0.006	−0.157	347
C10–C12 Food, beverages, tobacco	UV	1.86	0.31	0.006	−0.156	354
C13–C15 Textiles, apparel, leather	UV	2.88	0.15	0.009	−0.162	345
C16 Wood	UV	1.21	0.24	0.006	−0.158	397
C17 Paper	UV	1.80	0.41	0.019	−0.158	335
C19 Refined petroleum	UV	3.38	0.85	0.089	−0.151	133
C20 Chemicals	UV	4.31	0.46	0.077	−0.162	281
C21 Pharmaceuticals	UV	0.51	−0.07	0.000	−0.159	360
C22 Rubber and plastics	UV	1.68	0.14	0.002	−0.159	376
C23 Non-metallic minerals	UV	1.55	0.16	0.008	−0.156	345
C24 Basic metals	UV	4.17	0.47	0.213	−0.156	74
C25 Fabricated metals	UV	1.32	0.17	0.003	−0.159	376
C26 Electronics	UV	1.17	0.28	−0.006	−0.162	364
C27 Electrical equipment	UV	1.33	0.11	−0.002	−0.159	358
C28 Machinery	UV	0.93	0.01	−0.002	−0.159	360
C29 Motor vehicles	UV	0.82	0.09	−0.002	−0.158	386
C30 Other transport equipment	UV	0.42	−0.01	0.000	−0.159	351
C31–C32 Furniture, other goods	UV	0.49	0.10	−0.006	−0.158	410
D35 Electricity and gas	WB	2.50	2.32	0.141	−0.171	205
E37–E39 Waste and scrap	UV	2.25	0.52	0.215	−0.164	36
All products (baseline)		38.85		1.000	−0.159	352
Energy products		6.74		0.368	−0.161	52
Waste and scrap, basic metals		6.42		0.428	−0.161	14

Notes: The goods of Z^f_{jt}, eq. (23). Price: UV, chained index of the US dollar unit values of Türkiye's imports in UN Comtrade, from 4,840 six-digit HS codes with 2013 imports of 208.1 billion US dollars (80% of all imports); WB, World Bank prices as annual averages, with gas weighted by 0.803 and coal by 0.197 for electricity and gas. Weight: share in the variable cost of the sector's input–output column, eq. (24), mean over the 63 sectors; crude petroleum enters only C19, outside the sample. Log change: $\ln P^f_{p,2022} - \ln P^f_{p,2020}$. Rotemberg weight: α_p in the 2SLS estimate of eq. (19) (Goldsmith-Pinkham et al., 2020); the weights sum to one, the negative ones to −0.018, and the products whose own just-identified estimate is negative carry 0.993. The Herfindahl index of the products' shares in the instrument, proportional to $\sum_j m_j \omega_{jp}$, is 0.065, an effective number of 15.3 products (Borusyak et al., 2022); since world prices are correlated over time, the years of a product are not counted as separate shocks. Last two columns: eq. (19) with the product or group dropped from the instrument, and its cluster-robust first-stage F. Energy products: coal, crude petroleum, natural gas, refined petroleum, electricity and gas.

Table OA2. Real interest rate control and long difference

	Eq. (19)		Eq. (20)		
	β	F	β^f	β^e	N/G
(1) Baseline	−0.159***	352	−0.184***	−0.212***	719/63
	(0.027) [0.006]		(0.031) [0.005]	(0.056) [0.015]	
(2) $m_j\times$ real interest rate	−0.168***	84	−0.161***	−0.188***	719/63
	(0.048) [0.026]		(0.031) [0.006]	(0.059) [0.044]	
(3) Long difference, 2017–2022	−0.121***	80	−0.128	−0.343	56/56
	(0.035) [0.005]		(0.087) [0.122]	(0.274) [0.241]	

Notes: 2SLS estimates of eqs. (19) and (20). Row (1) is column (1) of Tables 2 and 3. Row (2) adds m_j times the real interest rate, the commercial loan rate less expected inflation. Row (3) uses the changes over 2017–2022, one observation per sector, with a constant; it starts in 2017 because the real exchange rate ended 2022 at its 2018 level, which would leave its term in eq. (20) almost without variation. F: cluster-robust first-stage F of β. Standard errors clustered by sector in parentheses; stars and brackets as in Table 2.

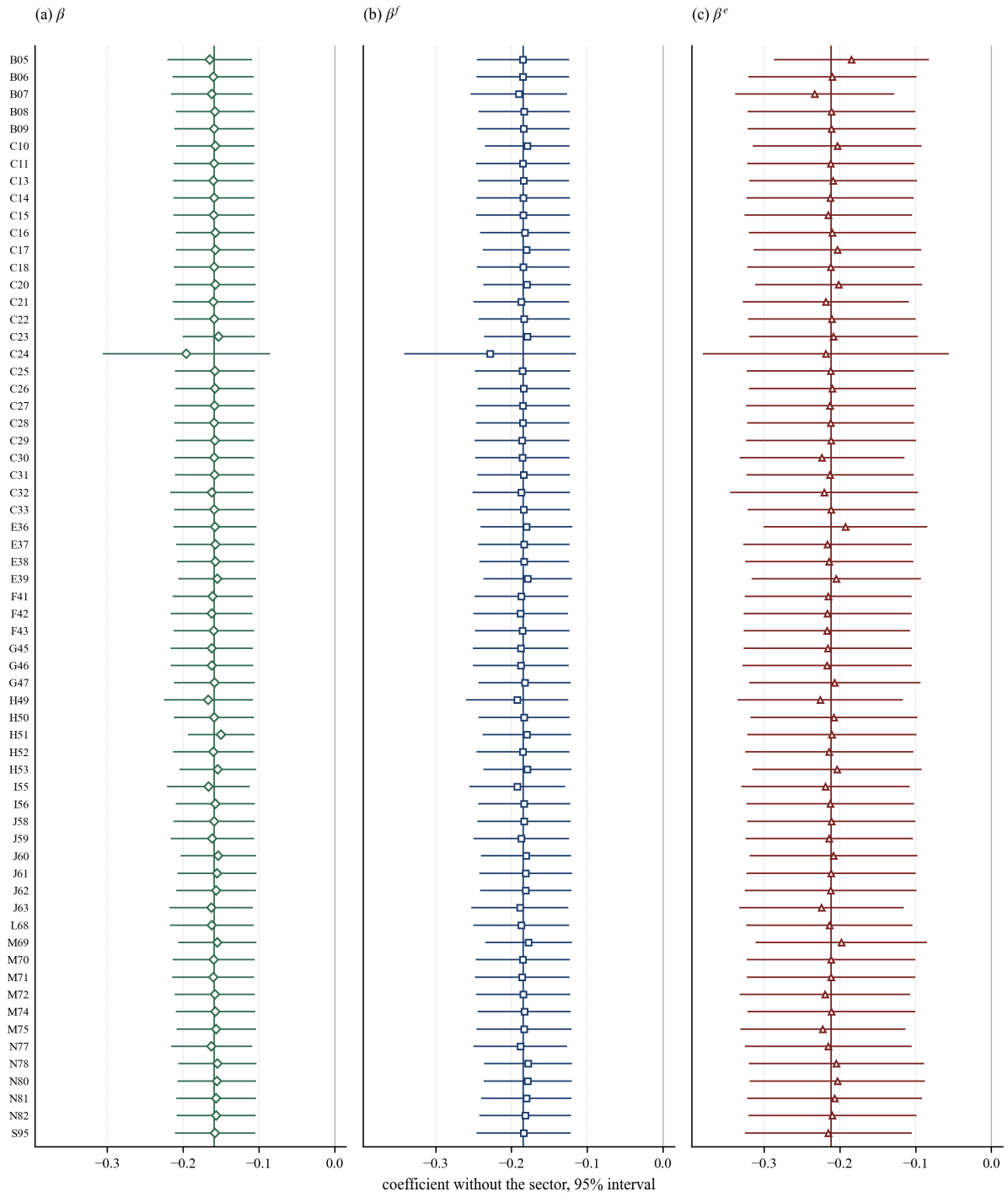


Figure OA1. Estimates without one sector at a time

Notes: Each row drops the sector named on the left: (a) β of eq. (19); (b) β^f and (c) β^e of eq. (20). Markers: estimates; lines: 95% intervals, clustered by sector; vertical line: the estimate on all 63 sectors.

Table OA3. Cumulative supplier price responses, with and without electricity and gas

	Month 0	Month 1	Month 2	Month 3	Month 12
Panel A: 21 supplier products					
Direct, ϕ_d	1.118***	1.207***	1.261***	1.294***	0.179
	(0.127)	(0.189)	(0.266)	(0.377)	(0.452)
Network, ϕ_n	1.358***	1.755***	2.600***	2.313*	3.754*
	(0.336)	(0.538)	(0.941)	(1.329)	(2.212)
Panel B: Without electricity and gas (D35), 20 products					
Direct, ϕ_d	1.134***	1.198***	1.068***	1.007***	−0.344*
	(0.165)	(0.235)	(0.221)	(0.241)	(0.193)
Network, ϕ_n	1.518*	1.666	0.759	−0.410	−1.216*
	(0.846)	(1.135)	(0.926)	(1.079)	(0.681)

Notes: Cumulative coefficients of eq. (25), split by route, with twelve monthly lags and product and month effects, 2013M1–2024M12; 3,024 product-months in Panel A and 2,880 in Panel B. Months 2 and 12 of Panel A are columns (4) and (6) of Table 4; month 0 differs from column (2), which has no lags. Standard errors clustered by product in parentheses; stars as in Table 2.

OA.5 The lagged coefficient of Table 5

Table OA4 splits the coefficients of column (3) of Table 5 into parts that add up exactly. The partial reversal in the following year comes mostly from the stock term and from personnel costs catching up; with value added as published, the lagged coefficient is small and not significant.

Table OA4. The coefficients of Table 5, column (3), by component

	Year of the change, β_0	Following year, β_1	Two-year sum
(1) Labor share, $\ln s_L$	−0.257***	0.111**	−0.146***
	(0.031)	(0.045)	(0.039)
(2) Stock term, $\ln(VA^{pub}/VA)$	−0.095**	0.062***	−0.033
	(0.038)	(0.016)	(0.038)
(3) Labor share with published value added, $\ln s_L^{pub}$	−0.162***	0.049	−0.113***
	(0.051)	(0.051)	(0.039)
(4) Personnel costs, $\ln W$	0.009	0.042**	0.051
	(0.028)	(0.018)	(0.039)
(5) Wage, $\ln w$	0.011	0.028*	0.039*
	(0.011)	(0.015)	(0.021)
(6) Employment, $\ln L$	−0.002	0.014	0.012
	(0.024)	(0.012)	(0.033)
(7) Published value added, $-\ln VA^{pub}$	−0.171***	0.007	−0.164***
	(0.060)	(0.061)	(0.060)

Notes: 2SLS estimates of eq. (26), as in column (3) of Table 5, with each component as the dependent variable, on the 643 sector-years of 2014–2024 with positive published value added; Sanderson–Windmeijer F 135 and 293 for the current and the lagged regressor. Rows (1) = (2) + (3), (3) = (4) + (7) and (4) = (5) + (6). On all 651 sector-years the lagged coefficient is 0.113 (Table 5). Standard errors clustered by sector in parentheses; stars as in Table 2.

OA.6 Local projections

Table OA5 gives the estimates behind Figure 2. For $h = 0, \ldots, 3$,

$$\ln s_{L,j,t+h} - \ln s_{L,j,t-1} = \beta_h \, m_j \ln\left(\frac{P_{m,jt}}{C_{jt}}\right) + \gamma_h \, m_j \ln\left(\frac{P_{m,j,t-1}}{C_{j,t-1}}\right) + \delta_j^h + \delta_t^h + u_{j,t+h}, \qquad \text{(OA.1)}$$

where C_{jt} is the wage and δ_j^h and δ_t^h are sector and year effects. The second regressor holds the previous year's price fixed, so β_h is the response to a rise in year t, and both regressors are instrumented by $m_j Z^f$ of the same year. In the year before a rise in the relative price, the labor share was already falling, although this is significant only at the 10% level under the wild cluster bootstrap (Table OA5, notes). The last row therefore controls for the change in the labor share in the previous year; the fall on impact and the rise after two years remain.

Table OA5. Local projections of the labor share

	$\ln s_{L,jt}$	$\ln s_{L,j,t+h} - \ln s_{L,j,t-1}$			
	(1) Static	(2) $h = 0$	(3) $h = 1$	(4) $h = 2$	(5) $h = 3$
$m_j \ln(P_{m,jt}/w_{jt})$	-0.208^{***}	-0.205^{***}	-0.031	0.283^{***}	0.159^{***}
	(0.035) [0.024]	(0.038) [0.030]	(0.073) [0.715]	(0.035) [0.006]	(0.040) [0.115]
Static prediction		-0.208	-0.285	0.108	0.163
Path of $m_j \ln(P_m/w)$		1	1.373^{***}	-0.520^{***}	-0.786^{***}
			(0.147)	(0.142)	(0.130)
Controlling for $\Delta \ln s_{L,j,t-1}$		-0.258^{***}	-0.125^{*}	0.175^{***}	0.055
		(0.047) [0.024]	(0.064) [0.069]	(0.027) [0.014]	(0.039) [0.327]

Notes: Column (1): eq. (19) on the observations of columns (2)–(5). Columns (2)–(5): eq. (OA.1) with start years 2014–2021 and the same 447 observations of 59 sectors at every horizon (442 with the control). First-stage F of Sanderson and Windmeijer (2016): 355 and 129 (398 and 138 with the control), 494 in column (1). Path: the response of $m_j \ln(P_{m,jt}/w_{jt})$ itself; static prediction: column (1) times the path. Per unit of the path, the response averages -0.256 over $h = 1$ to 3, against -0.205 at $h = 0$; the difference, -0.051 (0.055), is not significant. With $\ln s_{L,j,t-2} - \ln s_{L,j,t-1}$ on the left, the coefficient is 0.175 (0.038) [0.062]. Standard errors clustered by sector in parentheses; stars and brackets as in Table 2.

OA.7 Contributions to the labor share

Sector contributions from eq. (28) and from its counterpart for eq. (19) are converted into pp with the previous-year labor share, weighted by previous-year value added and summed over the years of a period. For the national accounts labor share, they are averaged within sections with previous-year personnel costs as weights and converted with section value added and labor shares; they are zero for sections A, D, P, Q and R.

In 2018–2022 the labor share of the 53 sectors of Table 7 fell from 41.96% to 30.54%, mostly within sectors, and the year effects of eq. (19), which the contributions leave out, fell by 0.123 log points. The 95% interval of the two-channel total is $[-2.18, -1.07]$ pp. Split by the shocks themselves, in the reduced form with $m_j Z^f_{jt}$ in place of the dollar price term, higher world prices lowered the labor share by 1.73 pp and the real exchange rate by 0.29 pp. The real exchange rate ended 2022 close to its 2018 level, so its part reflects only the timing of its rise in 2019–2020 and its fall in 2021–2022 (Section 4.5). This split need not add up to the two-channel total, since the dollar price of materials also moves with domestic shocks.

OA.8 The elasticity of substitution

Table OA6 gives estimates of eq. (27) beyond Table 6. The estimate is below one in levels, in one- to five-year differences and with materials measured as intermediate consumption, but it is imprecise. Most of the information comes from 2021–2022 and from energy prices: without these years, or without energy products in the instrument, the estimate stays below one but no longer differs significantly from one.

Table OA6. The elasticity of substitution between materials and primary inputs

	$\hat{\sigma}$	SE	95% interval	p, $\sigma = 1$	F	Obs.
Levels (Table 6, Panel C)	0.139	0.297	[−0.44, 0.72]	0.007	53	585
One-year differences	0.025	0.255	[−0.48, 0.53]	0.004	86	480
Two-year differences	−0.027	0.222	[−0.46, 0.41]	0.000	87	423
Three-year differences	−0.033	0.228	[−0.48, 0.41]	0.001	54	377
Four-year differences	0.120	0.306	[−0.48, 0.72]	0.009	23	330
Five-year differences	0.136	0.341	[−0.53, 0.80]	0.006	37	287
Long difference, 2018–2022	0.002	0.163	[−0.32, 0.32]	0.000	61	43
Materials as intermediate consumption	0.232	0.322	[−0.40, 0.86]	0.023	32	706
Instrument without energy products	−0.171	0.646	[−1.44, 1.09]	0.057	11	585
Without 2021–2022	0.327	0.476	[−0.61, 1.26]	0.144	16	488

Notes: Eq. (27) by 2SLS with the unscaled instrument Z^f_{jt} of eq. (23) and sector and year effects; $\hat{\sigma}$ is one minus the coefficient of $\ln(P_{m,jt}/w_{jt})$. Differences: overlapping changes over one to five years, with year effects and the change of Z^f_{jt} over the same years as the instrument; the long difference has one observation per sector and a constant. Intermediate consumption: materials adjusted for the change in stocks, the measure of materials in m_j. Energy products: coal, crude petroleum, natural gas, refined petroleum, and electricity and gas (Table OA1). Standard errors clustered by sector, and the 95% interval from them; p: wild cluster bootstrap test of $\sigma = 1$, as in Table 6. F: cluster-robust first-stage F. Obs.: sector-years (sectors for the long difference).

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